Assem Omar

Full-Stack .NET Software Engineer | C#, ASP.NET Core, Angular, SQL Server
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Professional Summary

Full-Stack Software Engineer specializing in **C#**, **ASP.NET Core**, **Angular**, and **SQL Server**, with hands-on experience developing enterprise web applications and production-ready software solutions. Experienced in building end-to-end business applications, implementing backend services and RESTful APIs, and integrating frontend interfaces with business logic. Strong foundation in **Clean Architecture**, **SOLID principles**, database design, and **Agile** software development, complemented by practical experience delivering solutions for real-world clients. B.Sc. in Computer Science (GPA: 3.5/4.0, ranked 5th in department) and graduate of ITI's 9-month Full-Stack .NET Professional Training Program.

Professional Experience

Excellent Protection Company — Full-Stack Software Engineer Cairo, Egypt
Full-time *Jun 2026 – Present*

- Designed and developed a full-stack HR management application using Angular and ASP.NET Core Web API, backed by SQL Server and Entity Framework Core, automating employee overtime, break-time tracking, and attendance calculations.
- Developed business logic and application workflows to translate HR requirements into functional software features, improving the automation of internal operations
- Integrated frontend components with backend services and implemented reusable application components, validation, and business rules
- Collaborated with business stakeholders in an **Agile** environment to analyze requirements, deliver incremental features, and maintain application functionality

Arab River Medical — Full-Stack API Developer Cairo, Egypt
Freelance Client Project | *armegypt.com* *2026*

- Designed and shipped a production-ready 3-tier **ASP.NET Core Web API (.NET 10)** with 15+ endpoints covering product catalogs, lead management, partner portals, and geo-based visitor analytics
- Implemented background jobs (Hangfire), SMTP email (MailKit), Geo-IP tracking, rate limiting, FluentValidation, and AutoMapper; containerized with **Docker** and prepared for cloud deployment
- Documented the API with Swagger/OpenAPI and kept a clean separation of concerns for maintainability

TranslateXable — Backend Developer Cairo, Egypt
Freelance Client Project | *translexable.io* *2026*

- Delivered a production backend for a live paying client: credit-based translation system, **Paymob** payment gateway integration, **JWT** authentication, and role-based access control (RBAC)
- Built admin, blog, and translation-history modules with full Swagger documentation; Dockerized for cloud deployment and client handoff
- Worked directly with the client through iterative feedback cycles as requirements evolved

Education

Information Technology Institute (ITI) ↗ Menofia, Egypt
9-Month Full-Stack .NET Professional Program (BI-Infused CRM Track) *Oct 2025 – Jun 2026*

- Professional training in ASP.NET Core, C#, SQL Server, enterprise application design, software architecture, and production engineering practices
- Team Project — NileChain | Live Demo: Full-stack agri-supply-chain marketplace

Benha University — Faculty of Computer Science & AI ↗ Benha, Egypt
B.Sc. in Computer Science — GPA: 3.5 / 4.0 (Very Good) *Oct 2021 – Jun 2025*

- Ranked 5th in the Department of Information Systems
- Graduation Project — TravelNest | GitHub: Led backend development in a 6-person team; designed 30+ REST APIs for authentication, bookings, and trip management with AI-powered recommendations

Projects

NileChain — AI-Powered Marketplace | *ASP.NET Core 10, Angular 21, EF Core, SQL Server, Docker* | GitHub

- Team project (ITI): AI-powered marketplace connecting Egyptian farms and factories through AI crop matching, digital contracts, escrow milestone payments, and wallets, built with **Clean Architecture** (API / Application / Domain / Infrastructure)
- Built Farm/Factory role-based profiles and dashboards (Chart.js), the JWT authentication flow, and Haversine-based farm matching end-to-end across backend and frontend; 12+ merged PRs in a 3-developer team

Sport Club Management System | *ASP.NET Core API, Angular, TypeScript, EF Core, SQL Server, JWT* | GitHub

- Full-stack, multi-role (Admin/Coach/Member) application for subscriptions, pitch bookings, and team/coach management, integrating an Angular UI with a C# business-logic layer
- Designed a normalized SQL Server schema with EF Core, Generic Repository, Unit of Work, and AutoMapper; built Angular role-based route guards, reactive forms, and a JWT HTTP interceptor

School Management System API | *ASP.NET Core, Clean Architecture, CQRS, EF Core, JWT, xUnit* | GitHub

- Architected a REST API with 30+ endpoints using 6-layer Clean Architecture and CQRS with MediatR; secured with JWT + RBAC and multi-language localization
- Wrote xUnit unit tests to validate service-layer logic and ensure reliability

AdvancedChat — Real-Time Chat App | *ASP.NET Core MVC, SignalR, EF Core, SQL Server, Identity* | GitHub

- Built a real-time platform with public, private, and group chat and persistent message history in SQL Server; implemented ASP.NET Identity authentication, SignalR-based online/offline presence, and role-based group access

RAG System — Document Q&A | *Python, Node.js, OpenAI Embeddings, ChromaDB* | GitHub

- Built a full RAG pipeline (ingestion, chunking, embeddings, vector search over 100+ chunks) exposed via a REST `/ask` endpoint with context-restricted LLM answers and a browser UI

Internships

DEPI — MERN Stack Web Development Intern ↗ Nasr-City, Egypt
Training Program | *GitHub* *Jun 2024 – Dec 2024*

- Built full-stack applications (React, Node.js, Express, MongoDB) with REST API integration, including dashboards, task-management tools, e-commerce features, and a Gemini AI chatbot

Raya Information Technology — RPA Automation Engineer Intern ↗ Maadi, Egypt
Internship *Jul 2023 – Sep 2023*

- Designed UiPath Orchestrator workflows for data extraction, validation, and report automation, reducing manual processing time by an estimated 40%

Technical Skills

Microsoft Technologies: C#, .NET 10, ASP.NET Core, ASP.NET Core Web API, ASP.NET Core MVC, Entity Framework Core, LINQ, ASP.NET Identity, SignalR
Frontend Development: Angular, React.js, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, Bootstrap
Database Development: Microsoft SQL Server, MySQL, PostgreSQL, MongoDB, Database Design, Relational Modeling, Query Optimization
Software Engineering: Object-Oriented Programming, SOLID Principles, Clean Architecture, Design Patterns, CQRS, Repository Pattern, Unit of Work, RESTful API Design
Cloud & DevOps: Microsoft Azure (Basic Knowledge), Docker, Git, GitHub
Testing & API Tools: xUnit, Unit Testing, Swagger/OpenAPI, Postman
Development Methodologies: Agile, Requirements Analysis, Iterative Development, Stakeholder Collaboration
Additional Technologies: Java, Python, Node.js, Spring Boot, RAG, OpenAI API, Gemini API

Languages

Arabic: Native **English**: Fluent